# Polya enumeration theorem

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**Cycle index polynomial**

Given a permutation group $G \leq S_n$, the cycle index polynomial of $G$ is defined as

$$Z_G(x_1, ..., x_n) = \frac{1}{|G|} \sum_{g \in G} x_1^{c_1(g)} ... x_n^{c_n(g)} = \frac{1}{|G|} \sum_{g \in G} \prod_{i=1}^{n} x_i^{c_i(g)},$$

where $c_i(g)$ is the number of cycles of length $i$ in the disjoint cycle decomposition of $g \in G \leq S_n$.

$Z_G(x_1, ..., x_n)$ is sometimes also written as $Z(G; x_1, ..., x_n)$.

Observe that cycle index polynomials are defined for permutation groups, not for actions directly. Given an action $\varphi : G \to \mathrm{Sym}(X)$, we can talk about the cycle index polynomial of $\mathrm{im}(\varphi) \leq S_{|X|}$, not of the action $\varphi$ itself.

The cycle index polynomial stores the information about *how* a permutation group group acts on its underlying set in an algebraically nice format. It separates out the elements of $G$ based on their cycle distribution (distribution of cycle sizes in the disjoint cycle decomposition of $g \in G$), and makes them independently manageable.

The division by $|G|$ pays tribute to the Burnside's lemma, of which this is a generalization. Observe that $c_1(g) = \mathrm{fix}(g)$, and so $\frac{\partial Z_G}{\partial x_1}|_{x_1,...,x_n=1} =$ number of orbits of the action of $G$ on $[n]$.

**Polya enumeration theorem**

We have $\Phi$ the set of "figures", and $w : \Phi \to \mathbb{N}$ (where $\mathbb{N}$ includes zero) is a function which assigns non-negative integer weights to each figure.

Let us have $G \leq S_n$, which acts naturally on $\Phi^n$, the set of $n$-tuples of figures.

Let us extend the function $w$ to $w' : \Phi^n \to \mathbb{N}$, which is the "total weight" of an $n$-tuple. Observe that total weight is invariant under the action of $G$ on $\Phi^n$.

Observe that the action of $G$ on $\Phi^n$ induces a "symmetry" on elements of $\Phi^n$; $x, y \in \Phi^n$ are isomorphic (or symmetrically identical) if they belong to the same orbit of the action.

Given $j \in \mathbb{N}$, we want to count the number of non-isomorphic (under the symmetry induced by the action) elements of $\Phi^n$ which have total weight $j$. Let us call this number $b_j$. The generating function of $b_j$ is defined as

$$b(t) = \sum_{i \geq 0} b_i t^i.$$

Now, let us define the figure counting series, $a(t)$ as

$$a(t) = \sum_{i \geq 0} a_i t^i,$$

where $a_i$ is the number of figures with weight $i$.

The **Polya enumeration theorem** states that

$$b(t) = Z_G(a(t), a(t^2), ..., a(t^n))$$

This boils down to showing that

$$b_j = [t^j]\left[\frac{1}{|G|}\sum_{g\in G} a(t)^{c_1(g)}a(t^2)^{c_2(g)}...a(t^n)^{c_n(g)}\right]$$

Let $\text{fix}_j(g)$ be defined as the number of elements of $\Phi^n$ which have total weight $j$ and are fixed by $g$. Applying Burnside's lemma on the action of $G$ on $\{a \in \Phi^n : w'(a) = j\}$, we get that

$$b_j = \frac{1}{|G|}\sum_{g\in G} \text{fix}_j(g).$$

Therefore, our goal simplifies down to showing that

$$\forall g \in G, \text{fix}_j(g) = [t^j]\left(a(t)^{c_1(g)}a(t^2)^{c_2(g)}...a(t^n)^{c_n(g)}\right)$$

Now, observe that for an $n$-tuple to be fixed by $g$, every cycle that appears in the disjoint cycle decomposition of $g$ has to choose a figure. And if a cycle of length $k$ choses a figure, then it contributes $k$ times the figure's weight to the total weight of the $n$-tuple. This is exactly what the above equation says, and so we are done.